

QINGYUAN, China

WMO: 542590

Lat: 42.0736N

Lon: 124.8660E

Elev: 230

StdP: 98.59

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-24.1	-29.3	0.3	-25.3	-27.3	0.3	-22.9	7.2	-6.3	6.3	-7.2	0.4	250	0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	32.0	21.8	30.7	21.7	29.3	21.2	24.7	29.1	23.9	27.8	23.1	26.7	3.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.5	18.8	26.7	22.7	17.9	26.0	22.0	17.2	25.4	76.5	29.2	73.3	27.8	70.3	26.8	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.2	5.2	DB	-29.4	34.3	2.4	1.7	-31.1	35.5	-32.5	36.5	-33.9	37.5	-35.6	38.7	(4)
(5)			WB	-29.5	25.7	2.4	1.0	-31.2	26.3	-32.7	26.9	-34.0	27.4	-35.8	28.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-13.7	-8.7	-0.4	8.4	15.4	20.8	23.9	22.6	16.6	8.3	-1.7	-11.0	(6)
(7)		DBStd	13.67	4.79	5.45	5.00	4.73	3.52	2.78	2.18	2.68	3.45	4.48	5.92	5.29	(7)
(8)		HDD10.0	2757	733	524	322	84	3	0	0	0	2	86	354	649	(8)
(9)		HDD18.3	4636	992	758	579	298	104	9	1	3	71	311	602	908	(9)
(10)		CDD10.0	1590	0	0	1	37	170	323	432	392	201	33	2	0	(10)
(11)		CDD18.3	427	0	0	0	1	12	83	174	137	20	0	0	0	(11)
(12)		CDH23.3	3611	0	0	0	27	245	776	1373	997	187	6	0	0	(12)
(13)		CDH26.7	1122	0	0	0	3	55	261	471	307	26	0	0	0	(13)
(14)	Wind	WSAvg	1.8	1.3	1.7	2.3	2.6	2.4	1.8	1.6	1.5	1.5	1.7	1.6	1.3	(14)
(15)	Precipitation	PrecAvg	811	8	9	17	47	68	116	195	188	77	47	26	11	(15)
(16)		PrecMax	1257	32	36	58	100	132	268	451	489	170	126	62	28	(16)
(17)		PrecMin	517	1	1	1	14	26	60	26	32	17	6	5	0	(17)
(18)		PrecStd	173	7	9	13	24	23	51	89	106	37	30	14	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	8.5	16.9	26.8	31.0	33.7	34.5	33.4	29.0	23.9	16.3	5.6	(19)
(20)			MCWB	0.1	3.0	8.6	13.7	17.3	19.3	24.0	23.2	20.6	15.1	9.8	2.4	(20)
(21)		2%	DB	-0.6	5.2	13.3	23.2	28.4	31.4	32.2	31.1	27.1	21.2	12.7	2.4	(21)
(22)			MCWB	-2.9	1.1	5.9	12.0	15.8	19.1	23.2	23.2	18.7	13.4	7.7	0.2	(22)
(23)		5%	DB	-2.8	3.3	10.8	20.8	26.3	29.6	30.8	29.8	25.5	18.8	10.1	0.1	(23)
(24)			MCWB	-4.8	-0.2	4.4	10.6	15.0	19.0	22.9	22.7	17.5	12.0	6.1	-1.9	(24)
(25)		10%	DB	-5.1	0.9	8.2	18.3	24.1	27.8	29.3	28.4	23.8	16.5	7.3	-2.1	(25)
(26)			MCWB	-6.8	-1.8	2.9	9.4	14.2	18.7	22.5	21.9	16.8	10.9	4.3	-3.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	3.8	9.1	15.5	19.2	22.2	26.1	25.6	22.1	16.4	10.6	2.8	(27)
(28)			MCDWB	2.5	6.9	16.1	24.1	27.1	27.7	31.6	30.0	27.4	21.9	14.9	4.6	(28)
(29)		2%	WB	-2.7	1.8	6.7	13.1	17.4	21.4	25.1	24.6	20.5	14.4	8.3	0.3	(29)
(30)			MCDWB	-0.8	4.8	12.0	20.8	25.1	26.8	29.9	28.8	24.4	19.4	11.8	2.1	(30)
(31)		5%	WB	-4.7	0.0	4.9	11.5	16.3	20.6	24.2	23.9	19.3	13.0	6.6	-1.7	(31)
(32)			MCDWB	-2.9	2.8	10.1	19.4	23.6	26.0	28.4	27.7	23.2	17.6	9.6	0.0	(32)
(33)		10%	WB	-6.9	-1.7	3.2	10.0	15.2	19.9	23.5	23.2	18.1	11.6	4.4	-3.8	(33)
(34)			MCDWB	-5.2	1.0	8.0	16.9	21.9	25.2	27.3	26.5	22.0	15.6	7.2	-2.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.8	13.7	11.9	13.6	13.5	11.5	9.2	9.7	13.0	13.5	11.3	12.4	(35)
(36)			MCDBR	15.0	14.7	15.6	18.1	18.0	15.5	12.1	11.9	15.0	16.6	14.5	13.4	(36)
(37)			MCWBR	12.7	10.6	8.9	8.7	6.9	5.2	4.3	4.6	6.8	8.7	9.5	11.0	(37)
(38)		5% WB	MCDBR	14.6	13.4	14.6	15.4	14.1	11.2	9.5	9.2	12.0	14.4	12.7	13.2	(38)
(39)			MCWBR	12.5	9.9	8.8	7.8	6.2	4.5	3.9	4.0	5.8	8.2	8.9	11.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.366	0.424	0.496	0.585	0.637	0.695	0.661	0.580	0.507	0.525	0.443	0.379	(40)	
(41)		taud	2.005	1.854	1.783	1.665	1.608	1.567	1.677	1.839	1.938	1.802	1.915	1.985	(41)	
(42)		Ebn at Noon	726	737	734	702	679	641	658	700	720	627	625	664	(42)	
(43)		Edh at Noon	118	161	196	237	257	269	239	197	166	169	129	111	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.39	3.31	4.35	4.93	5.56	5.17	4.70	4.65	4.37	3.23	2.19	1.92	(44)	
(45)		RadStd	0.16	0.22	0.24	0.34	0.45	0.53	0.41	0.41	0.31	0.21	0.25	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page